

CS Notes (Groups)

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Introduction

Creating Groups in Python

Permutations with SymPy

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└ Outline

Introduction

Creating Groups in Python

Permutations with SymPy

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└ Introduction

Introduction

Introduction

We will use **Python** in this course to perform mathematical computations. For the first part of the computer science course, we will look at how to use Python to work with groups. We will primarily interact with Python using the **Spyder** IDE.

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└ Introduction

└ Python

Python

We will use **Python** in this course to perform mathematical computations. For the first part of the computer science course, we will look at how to use Python to work with groups. We will primarily interact with Python using the **Spyder** IDE.

The fundamental objects we need from Python are:

- **lists**: [obj1, obj2, ...]
- **functions**: `def func(x): ...`

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└ Creating Groups in Python

Creating Groups in Python

Creating Groups in Python

A Simple Group in Python

Suppose we want to model the coins group of the previous class. Then we will need to specify the *objects* and the *actions* of the group. We can use lists to hold the objects. . .

```
# group from 1.1 has as objects arrangements of a penny and a
↪ dime
arrangement1 = ['penny', 'nickel']
arrangement2 = ['nickel', 'penny']
objects = [arrangement1, arrangement2]
```

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└ Creating Groups in Python

└ A Simple Group in Python

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A Simple Group in Python

...and functions to represent the actions:

```
# our action takes arrangement1 and turns it into arrangement2,  
↪ and  
# vice-versa  
def group_action(arrangement):  
    if arrangement[0] == 'penny':  
        return arrangement2  
    else:  
        return arrangement1  
  
# check that this does what we want it to  
for object in objects:  
    print(group_action(object))
```

['nickel', 'penny']

['penny', 'nickel']

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└ A Simple Group in Python

A Simple Group in Python

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['nickel', 'penny']  
['penny', 'nickel']
```


Example (Combining group actions using Python)

Find the result of performing the group action twice in a row using the code above.

Example (Words group again)

Implement the "three letter words" group in Python. Be careful to specify the objects and the actions!

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└ Creating Groups in Python

└ Group Actions in Python

There should be six objects and six actions that act on the objects!

Example (Combining group actions using Python)

Find the result of performing the group action twice in a row using the code above.

Example (Words group again)

Implement the "three letter words" group in Python. Be careful to specify the objects and the actions!

Recall that a symmetry group of a geometric figure is the set of actions that preserves the shape of that figure.

Example (Symmetry group of equilateral triangle)

Implement the symmetry group of the equilateral triangle in Python.

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└ Creating Groups in Python

└ Symmetry Groups in Python

This is more complicated than previous examples. Try working out examples by hand, then implementing in Python with a `for` loop.

Recall that a symmetry group of a geometric figure is the set of actions that preserves the shape of that figure.

Example (Symmetry group of equilateral triangle)

Implement the symmetry group of the equilateral triangle in Python.

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└─ Permutations with SymPy

Permutations with SymPy

Permutations with SymPy

- The SymPy library lets us handle permutations in cycle notation relatively easily.
- First, we need to import useful commands as follows:

```
from sympy.combinatorics import Permutation
from sympy.combinatorics import PermutationGroup
from sympy.combinatorics import SymmetricGroup
```

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└ Permutations with SymPy

└ Permutations in Python

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```

- We can now create permutations as follows:

```
R = Permutation(1, 2, 3)
```

Example (Rotation followed by flip)

Let R denote clockwise rotation of the equilateral triangle by 120° and let F denote the reflection that leaves vertex 1 unchanged. Using SymPy, find $R * F$ and $F * R$.

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└ Permutations with SymPy

└ Computations from S_3

Computations from S_3

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Example (Klein Four Group)

The symmetry group of a non-square rectangle is called the *Klein Four Group*. Create the elements of this group in Python using SymPy as above. Does the order of "multiplication" matter in this group?

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└ Permutations with SymPy

└ More Computations

More Computations

Example (Klein Four Group)

The symmetry group of a non-square rectangle is called the Klein Four Group. Create the elements of this group in Python using SymPy as above. Does the order of "multiplication" matter in this group?

Break into four groups; each group tackles two rows of the table

Example (Cayley table)

Find the Cayley table of the symmetry group of the square

- SymPy can easily compute the parity of a permutation:

```
R.parity() # = 0 since R is even
```

0

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└ Permutations with SymPy

└ Parity in SymPy

Parity in SymPy

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```
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Example (Determine existence of solution)

Consider the following 15-puzzle:

8	12	2	5
11	1	6	13
7	14	10	15
9	4	3	

Figure 1: A modified 15-puzzle

Is it possible to solve this puzzle? That is, do there exist moves that get the puzzle back into its solution state?

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- CS Notes (Groups)
 - └ Permutations with SymPy
 - └ Modified 15-Puzzle

Modified 15-Puzzle

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